

Adaptive masking: six options

Working note, 1 September 2026 — not part of the thesis

The problem this addresses

Measured 2026-09-01: the extractor applies the *same* transform regardless of how hard the trial is. Across easy-to-hard buckets ΔSIR moves only $+3.80 \rightarrow +4.32$ dB and ΔSAR only $-17.45 \rightarrow -21.41$ dB, while the word-error outcome swings from -4.2 to $+23.1$ points. It invents roughly 9 % of its output energy (absolute SAR $+10.34$ dB) and does so everywhere, which is a good trade on hard trials and a bad one on easy trials.

What the screening sweep found (1 September 2026)

The inference-time blend $\hat{S}_\alpha = \alpha\hat{S} + (1 - \alpha)X$ was swept over $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ on 103 two-speaker trials. SDR, SIR and SAR all moved perfectly monotonically, confirming the blend behaves as predicted.

Globally $\alpha = 1$ wins — there is no interior optimum, so a global shift toward gentler masking is the wrong direction. **But the optimum per difficulty bucket spans the entire range:**

difficulty	$\alpha=0$	0.25	0.5	0.75	$\alpha=1$
easy (<25 %)	9.6	11.2	10.3	11.7	14.2
medium	40.5	36.8	58.6	46.6	40.9
hard	81.9	80.6	82.7	79.1	73.4
very hard (>100 %)	137.4	133.8	131.4	140.1	113.6

Easy trials want *no filtering*; hard trials want *full filtering*. A single constant cannot be right, so **adaptation is what the data supports**. An oracle picking α per bucket reaches **56.9 %** against the model’s 59.1 % — about **2.2 points** available, with 52.5 % as an unreachable per-trial ceiling.

Caveat: the word-error curve is non-monotonic ($\alpha=0.5$ is worse than doing nothing) while the signal measures are not. The blend is provably linear, so this is $n=103$ noise plus transcriber nonlinearity. Trust the endpoints and the per-bucket pattern, not individual interior values.

Notation

$X[t, b]$ is the mixture in the time–frequency domain, frame t , band b ; $m[t, b]$ the complex mask the estimation module predicts; $h[t, b]$ the model’s internal features; e the enrolment. The current model is

$$\hat{S}[t, b] = m[t, b] X[t, b].$$

T frames, $B = 32$ bands, 7.19 M parameters, trained on 4,976 trials. **The model is data-limited, not capacity-limited** — train separation improved monotonically for all 18 epochs while held-out peaked at epoch 7. Any option that adds capacity makes the measured problem worse.

1. Global mix-back constant (no learning)

$$\hat{S}_\alpha = \alpha\hat{S} + (1 - \alpha)X, \quad \alpha \text{ fixed, chosen offline}$$

+ Zero training, zero new parameters, zero added latency: one multiply–add per sample, and output sample n needs only input sample n .

- + Every α comes from *one* forward pass, so a whole family of systems costs nothing but transcription.
- Not a fix. Trades away the benefit on hard trials to stop the harm on easy ones, using a constant that cannot be right everywhere.
- No gradient — α is not a model parameter, so the model learns nothing.

2. Per-frame learned gate

$$\alpha[t] = \sigma(w^\top h[t] + b), \quad \hat{S}[t] = \alpha[t] (m[t] X[t]) + (1 - \alpha[t]) X[t]$$

Gradient is immediate and needs no trick:

$$\frac{\partial \hat{S}[t]}{\partial \alpha[t]} = m[t] X[t] - X[t], \quad \frac{\partial \alpha[t]}{\partial w} = \alpha[t] (1 - \alpha[t]) h[t].$$

- + Roughly 1–2k parameters against 7.19M: capacity risk negligible.
- + Learns *when* to be gentle, which the global constant cannot.
- + Backbone untouched, so the 2026-08-28 architecture freeze survives.
- One decision for the whole spectrum at each frame.
- Still needs a retrain (~ 6.2 h per arm).

3. Per-frame, per-band learned gate

$$\alpha[t, b] = \sigma(w_b^\top h[t, b] + b_b), \quad \hat{S}[t, b] = \alpha[t, b] (m[t, b] X[t, b]) + (1 - \alpha[t, b]) X[t, b]$$

- + The band-split architecture already carries per-band features, so this is architecturally natural rather than bolted on.
- + Can be gentle in the high bands, where masking artefacts live, and aggressive in the low bands, where speech energy is. That is a distinction no scalar gate can express.
- + Still only $\sim B(|h_b| + 1)$ parameters — tens of thousands, not millions.
- Slightly more to defend in the write-up than a scalar.
- Same retrain cost.

4. Artefact weighting in the loss (beta) — no architecture change

The three components of the estimate are mutually orthogonal, so L_{pres} 's denominator is *already* the sum of the two piles:

$$\|\hat{s} - \alpha s\|^2 = \|e_{\text{interf}}\|^2 + \|e_{\text{artif}}\|^2$$

(verified numerically to six decimal places). They therefore cost the same per unit of energy. One ratio changes that:

$$L_{\text{sep}} = -10 \log_{10} \frac{\|\alpha s\|^2}{\|e_{\text{interf}}\|^2 + \beta \|e_{\text{artif}}\|^2 + \tau \|\alpha s\|^2}$$

- + $\beta = 1$ recovers L_{pres} *exactly*, so the control arm is the existing baseline and every prior run stays comparable.
- + No architecture change, no new parameters, no latency change.
- + Lets the model choose *where* in time and frequency to be gentle, which a global α cannot.
- + Unlike a separate artefact loss, pass-through can never win it: the interference term dominates.
- Shifts the global operating point; does *not* make the model adaptive.
- One new hyperparameter, and a 4-arm ablation is ~ 25 h of GPU time.

5. Soft-routing mixture of experts

$$g = \text{softmax}(f(X, e)) \in \mathbb{R}^K, \quad \hat{S} = \sum_{k=1}^K g_k (m_k X)$$

Differentiable without any trick, since softmax is smooth and the sum is linear:

$$\frac{\partial L}{\partial g_k} = \left\langle \frac{\partial L}{\partial \hat{S}}, m_k X \right\rangle.$$

- + No estimator bias, no sampling, no Gumbel trick.
- + Genuinely expresses “different treatment for different cases”.
- Replicating the whole estimator is fatal: $K=5$ would take the model from 7.19 M to 15.9 M, +122 %.
- + **Sharing the trunk changes the picture.** Measured: the estimator is 2.19 M of which **trunks** is 1.59 M (73 %), **mask_heads** 396 k and **res_heads** 198 k. Replicating only the mask heads costs +0.79 M at $K=3$, i.e. +11 % — and the separator (68 %) and trunks stay shared, so the experts are K read-outs of one representation rather than K models.
- All K masks are computed every frame, so runtime scales with K .
- Averaging masks can yield a mask worse than any individual one.

6. Variational mixture of experts (the version asked about)

Treat difficulty as a discrete latent $z \in \{1, \dots, K\}$ with prior $p(z)$ and posterior $q(z | X, e)$, and optimise the evidence lower bound

$$\mathcal{L} = \mathbb{E}_{q(z|X,e)}[L_{\text{recon}}(\hat{S}_z, S)] - \lambda \text{KL}(q(z | X, e) \| p(z)).$$

For small K the expectation is computed exactly by enumeration, so no sampling, no reparameterisation trick and no REINFORCE variance:

$$\mathcal{L} = \sum_{k=1}^K q_k L_{\text{recon}}(\hat{S}_k, S) - \lambda \sum_{k=1}^K q_k \log \frac{q_k}{p_k}.$$

- + **Mixes the losses, not the masks.** Each expert must be individually good on the cases assigned to it, avoiding the averaged-mask failure of soft routing.
- + The KL term prevents the gate collapsing onto a single expert, and λ controls how decisively it specialises.
- + Principled, citable, and a real architecture contribution.
- K full loss evaluations per step as well as K masks: the most expensive option by a wide margin.
- λ and $p(z)$ to choose on top of K .
- + The enumeration point is publishable in its own right: most treatments of discrete latents reach for Gumbel-Softmax or REINFORCE, and for K of 5–10 neither is needed.
- A new architecture is a different thesis from a new metric.

Summary

Option	Added Latency pa- ram- e- ters		Capacity risk	Build cost	Fixes non-adaptation
1. Global α	none	none	none	hours (no training)	no
2. Per-frame gate	$\sim 1\text{--}2\text{ k}$	none	tiny	1 retrain	partly
3. Per-band gate	$\sim 10\text{--}50\text{ k}$	none	small	1 retrain	yes
4. β in loss	none	none	none	4 retrains (25 h)	no
5. Soft MoE	$+11\%K \times \text{head}$ at $K=3$, shared trunk		moderate	1 retrain + tuning	yes
6. Variational MoE	$+11\%K \times \text{head}$ at $K=3$, shared trunk		moderate	largest	yes

Recommendation, after the sweep

The sweep has been run, and it re-orders the list.

Option 4 (β) is demoted. Globally $\alpha = 1$ already wins, so a global shift toward gentler masking is evidence-against rather than evidence-for. It stays a secondary arm — the sweep is an imperfect proxy, since a retrained model learns a *different* mask rather than a blended one — but it is no longer first.

Option 3 (per-band gate) is now the primary M5 experiment. The optimum varies from $\alpha = 0$ to $\alpha = 1$ across difficulty, so a single constant provably cannot be right. It costs tens of thousands of parameters against 7.19 M, leaves the backbone frozen, adds no latency, and there are 2.2 points of LCF-WER on the table.

And note the identity that links it to options 5 and 6:

$$\alpha(m \odot X) + (1 - \alpha)X = [\alpha m + (1 - \alpha)\mathbf{1}] \odot X$$

the per-band gate IS a two-expert mixture whose second expert is the identity mask. So it is not an alternative to the mixture-of-experts idea — it is that idea’s base case at near-zero cost, and the right way to test the routing hypothesis before paying for $K > 2$.

Options 5 and 6 are deferred, not rejected, and the capacity objection has been withdrawn. With a shared trunk, $K=3$ costs $+11\%$, which is defensible. What defers them is scope: six weeks to freeze and no live-model measurement taken. The sensible sequence is *build the $K=2$ gate, see how much of the 2.2 points it captures, and scale K only if it does*. If the gate saturates at $\alpha = 1$ everywhere, that is a cheap negative result that saves building the larger thing.